

- 11** A solid copper metal of mass 5.0 kg at an initial temperature T is placed with ice of mass 3.0 kg at 250 K. The copper and ice are in an insulated container of negligible mass. After thermal equilibrium is reached, the container has 2.2 kg of ice and 0.8 kg of liquid water.

Specific heat capacity of copper is $390 \text{ J kg}^{-1} \text{ K}^{-1}$.

Specific heat capacity of water is $4200 \text{ J kg}^{-1} \text{ K}^{-1}$.

Specific latent heat of fusion of ice is 340000 J kg^{-1} .

Specific heat capacity of ice is $2100 \text{ J kg}^{-1} \text{ K}^{-1}$.

What is the initial temperature T of the copper metal?

A 210 K

B 470 K

C 490 K

D 560 K

12 The temperature of one mole of oxygen gas is the same as that of one mole of nitrogen